

[Edit](#)

Create a Bot by Using the Bot Framework SDK

Challenge Overview

Understand the scenario

You are a Developer for Hexelo, an organization that would like to explore DALL-E 3's capabilities, focusing on image generation. In this Challenge Lab, you will create a bot by using the Bot Framework SDK. First, you will install the Bot Framework Emulator, and then you will create and enable a virtual environment. Next, you will create a bot, and then you will start the bot. Finally, you will start the Emulator and connect your bot.

Navigating the Challenge Lab

🔗 ► Understanding a Cloud Slice

🔗 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Install the Bot Framework Emulator.

Hints Enabled

No Yes

Note About Evolving Technology

Azure AI technologies are constantly evolving as Microsoft continues to update, refine, and improve its tools and interfaces. As a result, you may notice slight differences between the instructions in this lab and what you see on your screen. Rest assured, we are actively working to keep all tutorials up to date with the latest features and interfaces. We appreciate your patience and understanding, and we apologize for any inconvenience this may cause.

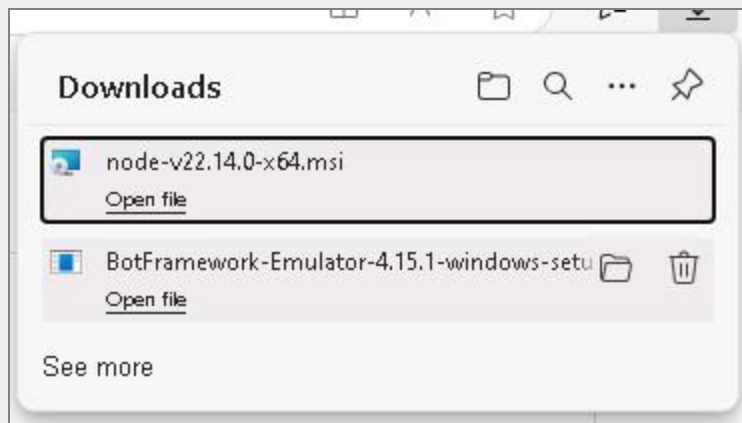
- Sign into the virtual machine as **Admin** by using  **PasswOrd!** as the password.
- Open **Microsoft Edge**, go to  <https://github.com/Microsoft/BotFramework-Emulator/releases/tag/v4.15.1>, and then download **BotFramework-Emulator-4.15.1-windows-setup.exe**.

💡 Expand this hint for guidance on downloading the Bot Framework Emulator.

- On the Windows taskbar, in Search, enter **Edge**, and then select **Microsoft Edge** to open a new browser.
- In the Address bar, enter <https://github.com/Microsoft/BotFramework-Emulator/releases/tag/v4.15.1>, and then select **Enter**.
- In Assets, select **BotFramework-Emulator-4.15.1-windows-setup.exe**.



- In the Downloads popup, select **Open file location**.

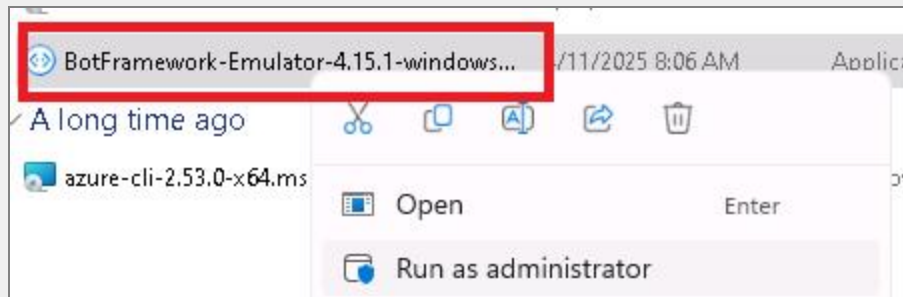


💡 Want to learn more? Review the documentation on [downloading the Bot Framework Emulator](#).

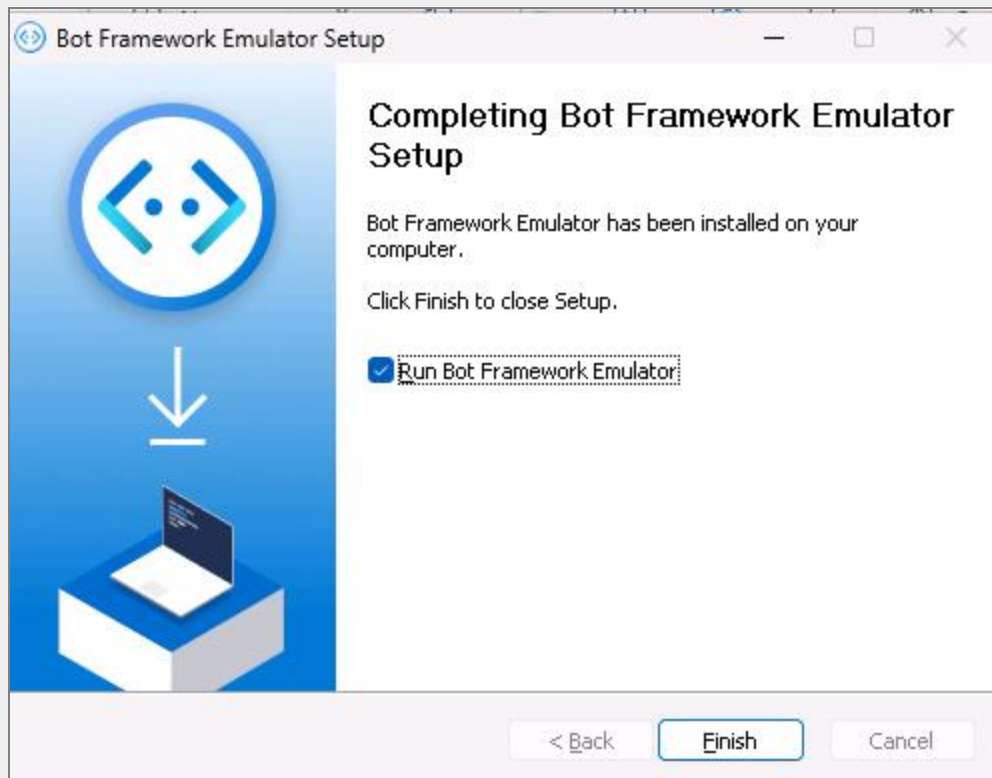
- Install the Bot Framework Emulator.

💡 Expand this hint for guidance on installing the Bot Framework Emulator.

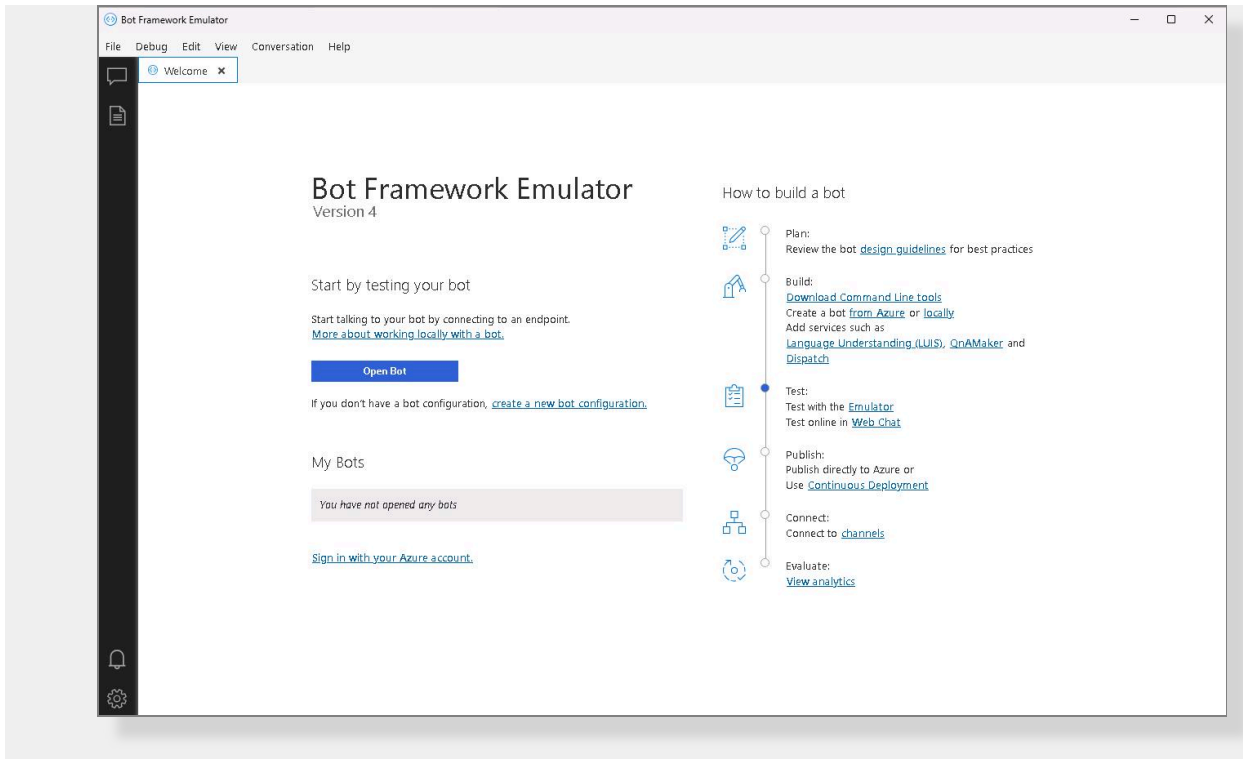
- In the Downloads folder, select **BotFramework-Emulator-4.15.1-windows-setup.exe**, right-click, and then select **Run as administrator**.



- When alerted *Do you want to allow this app to make changes to your device*, select **Yes**.
- On the Bot Framework Emulator Setup page, in Choose Installation Options, select **Only for me (Admin)**, and then select **Next**.
- On the Choose Install Location page, leave the default destination, and then select **Next**.
- Once installation has completed, select **Finish**.



- Minimize the Bot Framework Emulator once it launches.



- ❓ If you receive the following error, select **Retry**. This is a known bug in the Emulator Setup installation.



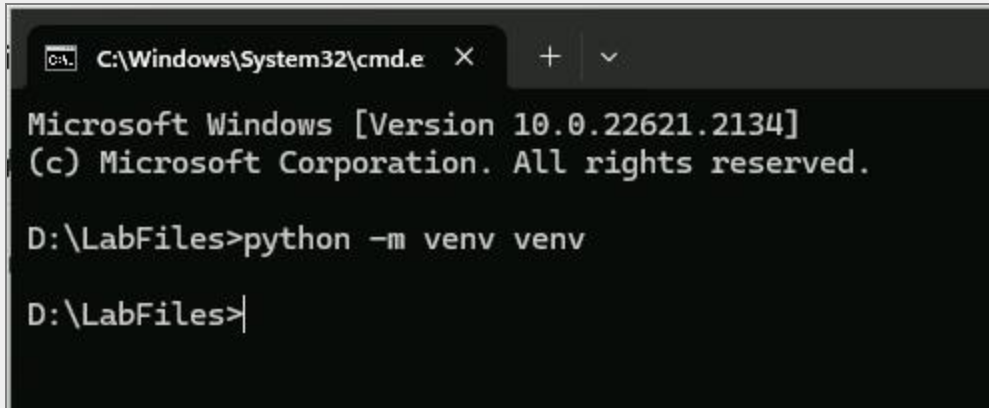
- Create a virtual environment named `T venv` in the `T D:\LabFiles` folder, and then activate the virtual environment.

💡 Expand this hint for guidance on creating and activating a virtual environment. ^

- In Windows Explorer, navigate to the `D:\LabFiles` folder, select the entire folder address in the address bar, enter `cmd`, and then select **Enter**.
- In the Command prompt, enter the following command to create the virtual environment, and then select **Enter**:

Bash

 `python -m venv venv`



```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.22621.2134]
(c) Microsoft Corporation. All rights reserved.

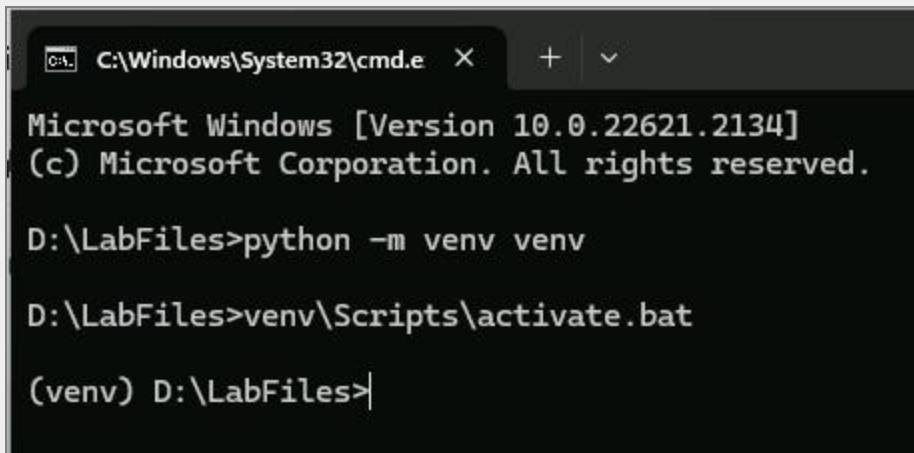
D:\LabFiles>python -m venv venv

D:\LabFiles>
```

- In the Command prompt, enter the following command to activate the virtual environment, and then select **Enter**:

Bash

 `venv\Scripts\activate.bat`



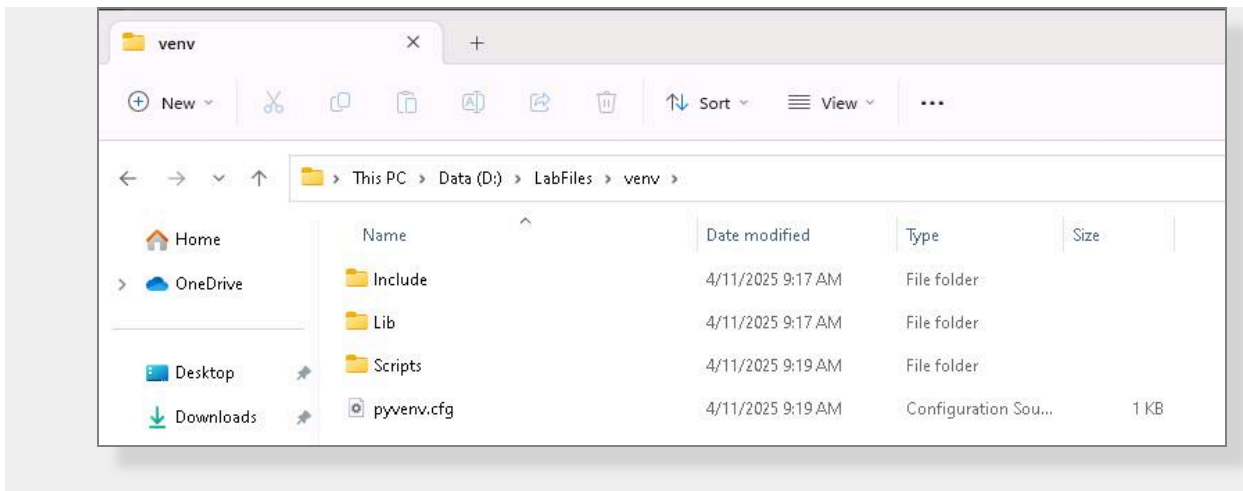
```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.22621.2134]
(c) Microsoft Corporation. All rights reserved.

D:\LabFiles>python -m venv venv

D:\LabFiles>venv\Scripts\activate.bat

(venv) D:\LabFiles>
```

- In the `D:\LabFiles` folder, open the new **venv** folder, and review its contents.

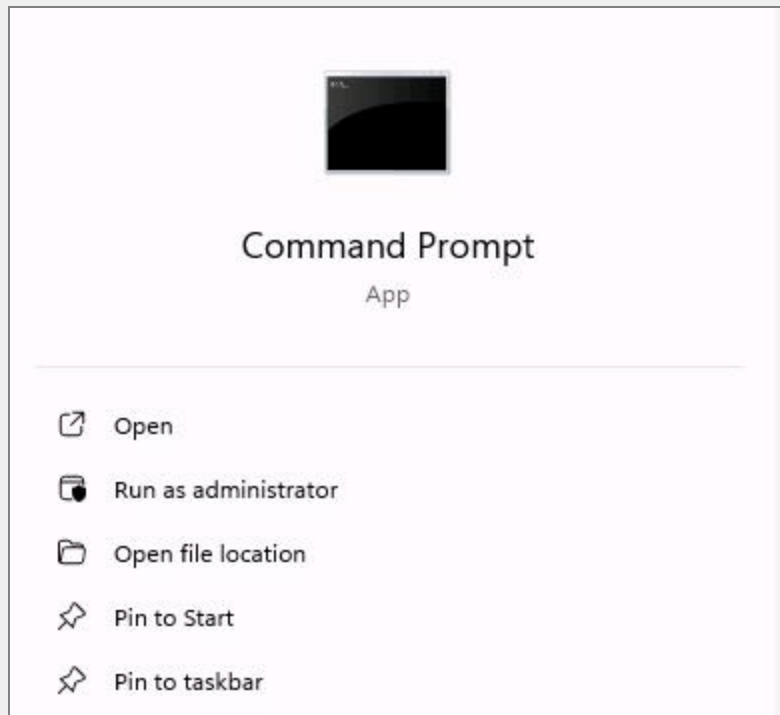


 Want to learn more? Review the documentation on [creating and activating a virtual environment](#).

- Install the following Python packages by running the following `pip install` commands as an Administrator:
 - `botbuilder-core`
 - `asyncio`
 - `aiohttp`
 - `cookiecutter==1.7.0`

💡 Expand this hint for guidance on installing the necessary python packages.

- On the Windows Desktop taskbar, in Search, enter  +cmd, and then select **Run as administrator**.



- In the Command prompt, enter the following commands to install the necessary packages, and then select **Enter** after each command:

Bash

 `pip install botbuilder-core`

```
C:\> Select Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.2134]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>pip install botbuilder-core
Collecting botbuilder-core
  Using cached botbuilder_core-4.16.2-py3-none-any.whl (115 kB)
Collecting botframework-streaming==4.16.2
  Using cached botframework_streaming-4.16.2-py3-none-any.whl (41 kB)
Collecting botframework-connector==4.16.2
  Using cached botframework_connector-4.16.2-py2.py3-none-any.whl (100 kB)
Collecting jsonpickle<1.5,>=1.2
  Using cached jsonpickle-1.4.2-py2.py3-none-any.whl (36 kB)
Collecting botbuilder-schema==4.16.2
  Using cached botbuilder_schema-4.16.2-py2.py3-none-any.whl (35 kB)
Collecting msrest==0.7.*
```

Bash

pip install asyncio

```
C:\Windows\System32>pip install asyncio
Collecting asyncio
  Using cached asyncio-3.4.3-py3-none-any.whl (101 kB)
Installing collected packages: asyncio
Successfully installed asyncio-3.4.3
WARNING: You are using pip version 21.2.3; however, version 25.0.1 is available.
You should consider upgrading via the 'C:\Python310\python.exe -m pip install --upgrade pip' command.

C:\Windows\System32>
```

Bash

pip install aiohttp

```
Administrator: Command Prompt

C:\Windows\System32>pip install aiohttp
Collecting aiohttp
  Using cached aiohttp-3.11.16-cp310-cp310-win_amd64.whl (442 kB)
Collecting async-timeout<6.0,>=4.0
  Using cached async_timeout-5.0.1-py3-none-any.whl (6.2 kB)
Collecting yarl<2.0,>=1.17.0
  Using cached yarl-1.19.0-cp310-cp310-win_amd64.whl (92 kB)
Collecting aiohappyeyeballs>=2.3.0
  Using cached aiohappyeyeballs-2.6.1-py3-none-any.whl (15 kB)
Collecting multidict<7.0,>=4.5
  Using cached multidict-6.4.3-cp310-cp310-win_amd64.whl (38 kB)
Collecting propcache>=0.2.0
  Using cached propcache-0.3.1-cp310-cp310-win_amd64.whl (45 kB)
Collecting attrs>=17.3.0
  Using cached attrs-25.3.0-py3-none-any.whl (63 kB)
Collecting aiosignal>=1.1.2
  Using cached aiosignal-1.3.2-py2.py3-none-any.whl (7.6 kB)
Collecting frozenlist>=1.1.1
  Using cached frozenlist-1.5.0-cp310-cp310-win_amd64.whl (51 kB)
Requirement already satisfied: typing-extensions>=4.1.0 in c:\python310\lib\site-packages (from multidict<7.0,>=4.5->aiohttp) (4.13.2)
Requirement already satisfied: idna>=2.0 in c:\python310\lib\site-packages (from yarl<2.0,>=1.17.0->aiohttp) (3.10)
Installing collected packages: propcache, multidict, frozenlist, yarl, attrs, async-timeout, aiosignal, aiohappyeyeballs, aiohttp
Successfully installed aiohappyeyeballs-2.6.1 aiohttp-3.11.16 aiosignal-1.3.2 async-timeout-5.0.1 attrs-25.3.0 frozenlist-1.5.0 multidict-6.4.3 propcache-0.3.1 yarl-1.19.0
WARNING: You are using pip version 21.2.3; however, version 25.0.1 is available.
You should consider upgrading via the 'C:\Python310\python.exe -m pip install --upgrade pip' command.

C:\Windows\System32>
```

Bash

pip install cookiecutter==1.7.0

```
C:\Windows\System32>pip install cookiecutter==1.7.0
Collecting cookiecutter==1.7.0
  Using cached cookiecutter-1.7.0-py2.py3-none-any.whl (40 kB)
Collecting poyo>=0.1.0
  Using cached poyo-0.5.0-py2.py3-none-any.whl (10 kB)
Collecting click>=7.0
  Using cached click-8.1.8-py3-none-any.whl (98 kB)
Collecting whichcraft>=0.4.0
  Using cached whichcraft-0.6.1-py2.py3-none-any.whl (5.2 kB)
Collecting jinja2-time>=0.1.0
  Using cached jinja2_time-0.2.0-py2.py3-none-any.whl (6.4 kB)
Requirement already satisfied: requests>=2.18.0 in c:\python310\lib\site-packages (from cookiecutter==1.7.0) (2.32.3)
Collecting future>=0.15.2
  Using cached future-1.0.0-py3-none-any.whl (491 kB)
Collecting jinja2>=2.7
  Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Collecting binaryornot>=0.2.0
  Using cached binaryornot-0.4.4-py2.py3-none-any.whl (9.0 kB)
Collecting chardet>=3.0.2
  Using cached chardet-5.2.0-py3-none-any.whl (199 kB)
```

- Close the Administrator: Command Prompt.



Want to learn more? Review the documentation on [installing the necessary python packages](#).

Check your work

Verify

Create a and start bot.

Hints Enabled

No Yes

- From the **D:\LabFiles** directory, run the following command to download the echo bot template and its dependencies:

Bash

 `cookiecutter https://github.com/microsoft/BotBuilder-Samples/releases/downl`



Expand this hint for guidance on downloading the echo bot template and its dependencies.



- In Windows Explorer, navigate the the **T** `D:\LabFiles` folder, select the entire folder address in the address bar, enter **T** `cmd`, and then select **Enter**.
- In the Command prompt, enter the following command to download the echo bot template and its dependencies, and then select **Enter**:

Bash

 `cookiecutter https://github.com/microsoft/BotBuilder-Samples/releases/do`

- When prompted to enter a **bot_name**, enter **T** `my_echo_bot`.

```
bot_name [my_chat_bot]: my_echo_bot
bot_description [Demonstrate the core capabilities of the Microsoft Bot Framework]: |
```

- When prompted to enter a **bot_description**, enter **T** `A bot that echoes back user responses`.

```
bot_name [my_chat_bot]: my_echo_bot
bot_description [Demonstrate the core capabilities of the Microsoft Bot Framework]: A bot that echoes back user response
s
D:\LabFiles>
```

 Want to learn more? Review the documentation on [downloading the echo bot template and its dependencies](#).

- Change directories to **my_echo_bot**.

 Expand this hint for guidance on changing directories. 

- In the Command prompt, enter the following command to change directories to  **my_echo_bot**, and then select **Enter**:

Bash

 `cd my_echo_bot`

```
D:\LabFiles>cd my_echo_bot
D:\LabFiles\my_echo_bot>|
```

- Install the dependencies for the **echo bot** template.

 Expand this hint for guidance on installing the dependencies for the echo bot template. 

- In the Command prompt, enter the following command to change directories to  **my_echo_bot**, and then select **Enter**:

Bash

 `pip install -r requirements.txt`

```
D:\LabFiles\my_echo_bot>pip install -r requirements.txt
Collecting botbuilder-integration-aiohttp==4.14.0
  Using cached botbuilder_integration_aiohttp-4.16.2-py3-none-any.whl (19 kB)
Requirement already satisfied: botframework-connector==4.16.2 in c:\python310\lib\site-packages (from botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (4.16.2)
Collecting aiohttp==3.10.5
  Using cached aiohttp-3.10.5-cp310-cp310-win_amd64.whl (379 kB)
Requirement already satisfied: botbuilder-core==4.16.2 in c:\python310\lib\site-packages (from botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (4.16.2)
Requirement already satisfied: yarl==1.8.1 in c:\python310\lib\site-packages (from botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (1.19.0)
Requirement already satisfied: botbuilder-schema==4.16.2 in c:\python310\lib\site-packages (from botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (4.16.2)
Requirement already satisfied: aiohappyeyeballs==2.3.0 in c:\python310\lib\site-packages (from aiohttp==3.10.5->botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (2.6.1)
Requirement already satisfied: multidict<7.0,>=4.5 in c:\python310\lib\site-packages (from aiohttp==3.10.5->botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (6.4.3)
Requirement already satisfied: frozenlist==1.3.1 in c:\python310\lib\site-packages (from aiohttp==3.10.5->botbuilder-integration-aiohttp==4.14.0->-r requirements.txt (line 1)) (1.3.1)
```



Want to learn more? Review the documentation on [installing the dependencies for the echo bot template](#).

- Start the bot.



Expand this hint for guidance on starting a bot.



- In the Command prompt, enter the following command to start **my_echo_bot**, and then select **Enter**:

Bash



python app.py

```
D:\LabFiles\my_echo_bot>python app.py
===== Running on http://localhost:3978 =====
(Press CTRL+C to quit)
```

- Copy the http address in the last line. You'll need it when you use the Emulator to interact with your bot.



Want to learn more? Review the documentation on [starting a bot](#).

Check your work

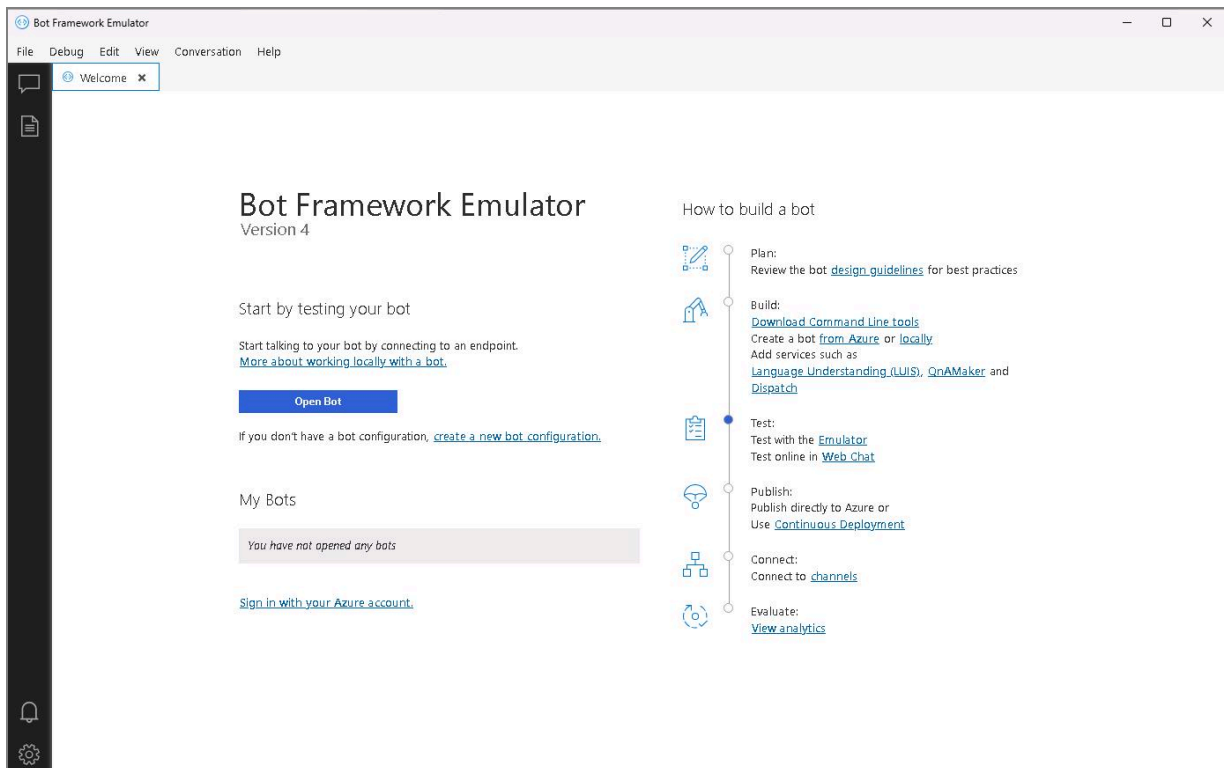
Verify

Start the Emulator and connect your bot.

Hints Enabled

No Yes

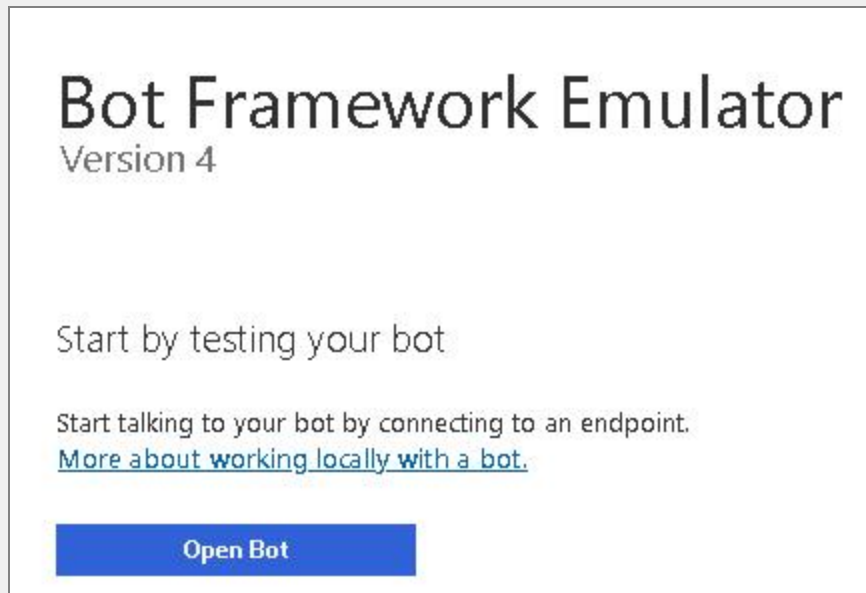
- If necessary, start the Bot Framework Emulator.



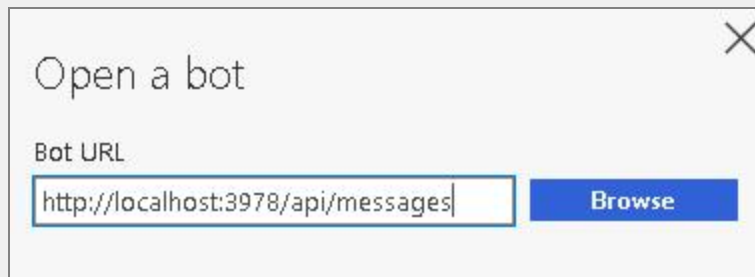
- Open the **my_echo_bot** in the Bot Framework Emulator.

💡 Expand this hint for guidance on opening a bot in the Bot Framework Emulator.

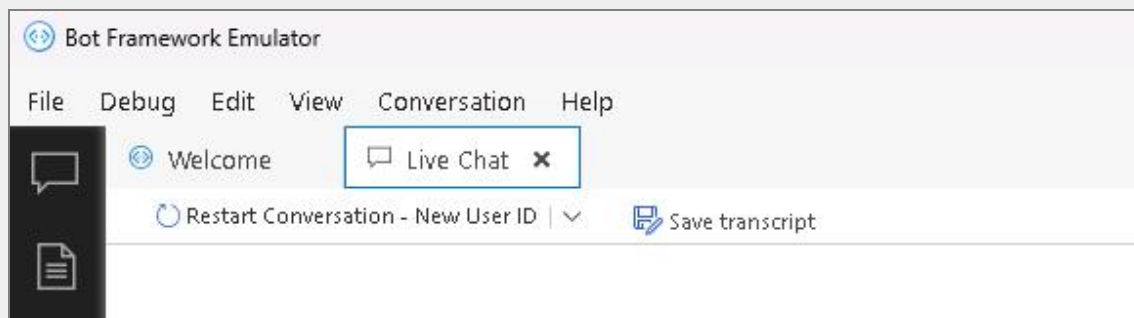
- On the Bot Framework Emulator Welcome page, in Start by testing your bot, select **Open Bot**.



- On the Open a bot dialog, in Bot URL, paste the copied URL from the previous step, append the path by adding `/api/messages` at the end, and then select **Connect**.



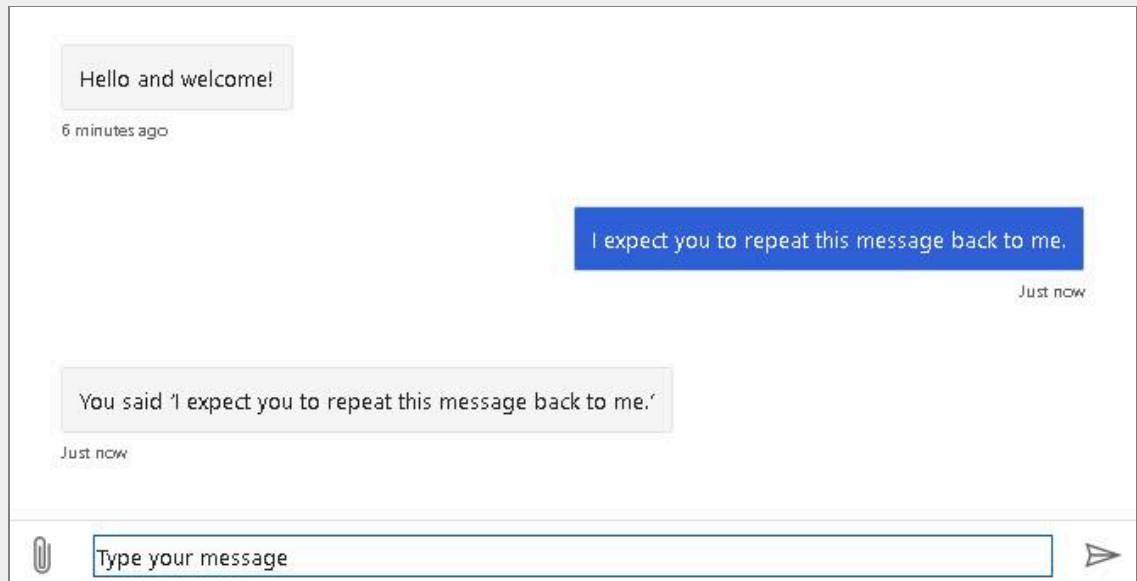
- You should now be on a new tab named **Live_Chat**.



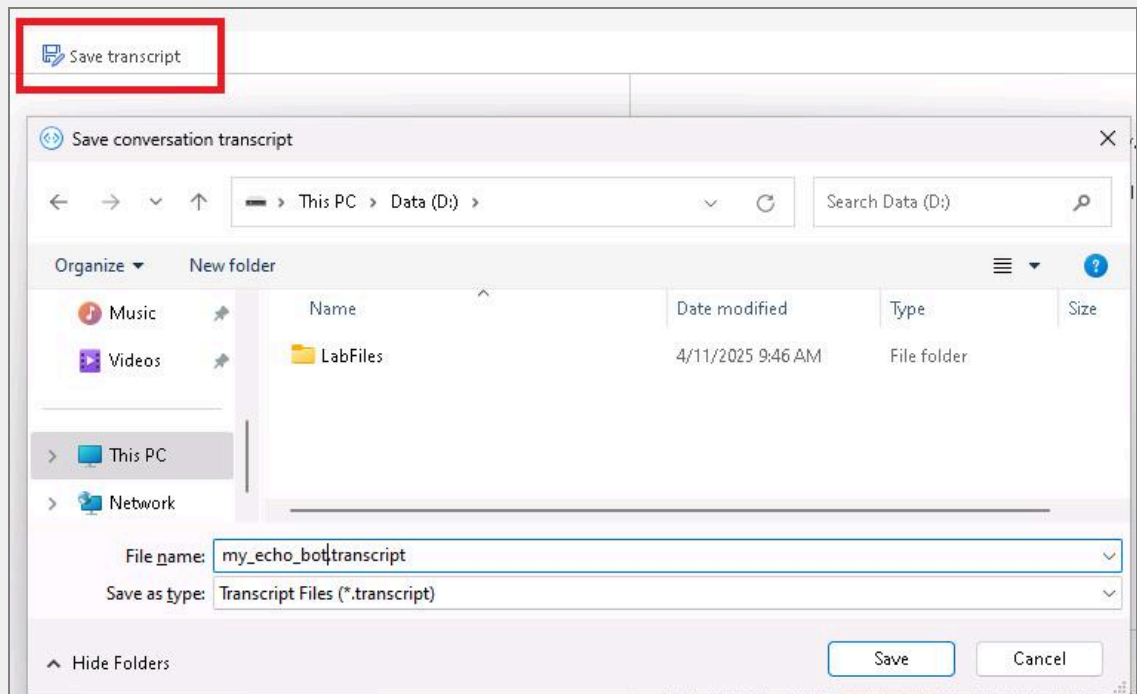
- Send the following message to your bot to see if the bot will respond back, and then save the transcript as: `I expect you to repeat this message back to me.`

💡 Expand this hint for guidance on sending a message to a bot.

- In the Bot Framework Emulator, in Type your message, enter **T** I expect you to repeat this message back to me., and then select **Send**.



- Select **Save transcript**, and then navigate to the **T** D:\LabFiles folder.
- On the Save conversation transcript window, in File name, enter **T** my_echo_bot.transcript, and then select **Save**.



💡 Want to learn more? Review the documentation on [sending a message to a bot](#).

Check your work

Verify

Summary

Congratulations, you have completed the **Create a Bot by Using the Bot Framework SDK** Challenge Lab.

You have accomplished the following:

- Installed the Bot Framework Emulator.
- Created and enabled a virtual environment.
- Created a bot.
- Started your bot.
- Started the Emulator and connected your bot.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Explore Document Processing in Azure AI's Document Intelligence Studio [Guided]
- Create a Bot by Using the Bot Framework SDK [Guided]
- Generate Insights from Images by Using Azure AI Vision Services [Guided]
- Build an AI Model in Azure AI Machine Learning Studio [Guided]
- Integrate Azure AI Immersive Reader Into Your Application by Using Its SDK [Guided]

- Summarize Text Automatically by Using Azure AI Language Services [Guided]
- Translate Text Between Languages by Using Azure AI Translator [Guided]
- Generate Image Descriptions from Text by Using Azure AI Services [Guided]
- Build an AI-Powered Search System by Using Azure AI Search [Guided]
- Analyze and Generate Text-Based Sentiment by Using Azure AI Services [Guided]

